

NEP BASED
SCHEME OF LEARNING
and
SYLLABUS FOR
B.TECH MAC
IV SEMESTER

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1. B. Tech (MAC) Semester-IV

B. Tech -SEMESTER IV														
Course Code	Type	Course	L	T	P	Credit s	Evaluation Scheme					Offering Dept.	Course Type	
							Theory			Practical			AICTE	NEP-2020
							CA	MS	ES	CA	ES			
CM CSC 4 01	CC	Theory of Automata & Formal Languages	3	1	0	4	30	20	50		-	CSE	Program Core	Discipline Specific
CM CSC 4 02	CC	Computer Architecture and Organization	3	1	0	4	30	20	50	-	-	CSE	Program Core	Discipline Specific
CM CSC 4 03	CC	Machine Learning	3	0	2	4	-	20	50	30	-	CSE	Program Core	Discipline Specific
CM CSC 4 04	CC	Design and Analysis of Algorithms	3	0	2	4	-	20	50	30	-	CSE	Program Core	Discipline Specific
CM ICC 4 05	CC	Optimization Techniques	3	1	0	4	30	20	50	-	-	ICE	Engg. Sc.	Inter-Disciplinary
VAXXxx x**	VAC	--	-	-	-	NIL	10 0	-	-	-	-	-	Mandatory	VAC
		22 Contact Hours**				20								
** Any one course from the approved list of VAC courses. Course code, L-T-P and evaluation scheme are given in the list of VAC courses. Actual teaching hours shall depend on LTP of the VAC course registered by the student.														

2. IV SEMESTER B.TECH (MAC) Detailed Syllabus of Core Courses

2.1 Theory of Automata & Formal Languages

A. OVERVIEW OF THE COURSE:

Name of Course	Theory of Automata & Formal Languages
Offering Department	COMPUTER SCIENCE & ENGINEERING

OVERVIEW:

The Course on **Theory of Automata & Formal Languages** is offered to the second-year students. Theory of Automata & Formal Languages course is to introduce the fundamental mathematical and computational principles that are the foundation of computer science. These include topics such as Turing machines, Automata, grammars and formal languages, decidability, halting problem, the $P = NP$ question and NP-Completeness reductions. A secondary objective is to address students' misconceptions that Theory of Computation is disconnected from state-of-the-art applications and today's open problems; once they complete a theory course, however, they understand that the fundamental theorems and proofs of theory provide the foundation material for all specialties of computer science. For example, Theory of Automata & Formal Languages introduces conceptual tools that practitioner's use in compiler and programming language. A final and related objective of a TOC course is to prepare students to be either well-rounded practitioners or potential candidates for computer science graduate programs. Determining whether a Theory of Automata & Formal Languages course prepares students for graduate school is straightforward: approximately 1/3 of the problems on the GATE test for computer science require an understanding of Theory of Automata & Formal Languages.

B. SYLLABUS

COURSE NO	TITLE OF THE COURSE	COURSE STRUCTURE	PRE-REQUISITE
CMCSC 401	Theory of Automata & Formal Languages	3L - 1T - 0P	None

APPROVED IN THE BOS HELD ON 19/11/2024

COURSE OUTCOMES (COs)

After completion of this course, the students are expected to be able to demonstrate the following knowledge, skills, and attitudes:

CO1: Understand formal languages, grammars, and automata, and analyze their relationships with computational models.

CO2: Apply concepts of finite automata and regular expressions to design and optimize regular language recognizers.

CO3: Construct context-free grammars and pushdown automata for context-free languages, and solve problems related to ambiguity and normal forms.

CO4: Analyze and design Turing machines to model complex computational problems and understand the limitations of computation.

CO5: Evaluate decidability and computational complexity of problems, distinguishing between tractable and intractable problems, and explore P, NP, and NP-hard classifications.

COURSE CONTENTS**Unit I**

Finite Automata and Regular Languages: Deterministic FA, Non-deterministic FA, Regular expressions, Finite Automaton with ϵ - moves, Regular Expression, Regular Languages and Kleene's theorem– Conversion of NFA to DFA, Equivalence of finite Automaton and regular expressions, Arden's Theorem. Myhill Nerode Theorem, Minimization of DFA, Pumping Lemma for Regular sets, Problems based on Pumping Lemma

Unit II

Context-Free Grammar and Languages: Grammar, Types of Grammar, Context Free Grammars and Languages, Derivations, Ambiguity, Relationship between derivation and derivation trees, Simplification of CFG, Elimination of Useless symbols - Unit productions - Null productions, Chomsky normal form (CNF), Greibach Normal form (GNF), Problems related to CNF and GNF

Unit III

Pushdown Automata and Context-Free Languages: Moves, Instantaneous descriptions, Deterministic/Non-deterministic, pushdown automata, Equivalence of Pushdown automata and CFL, pumping lemma for CFL, problems based on pumping Lemma, Conversion from CFG-to-PDA, Conversion from PDA-to-CFG

Unit IV

Turing Machines and Computability: Definition and Types of Turing machines, Computable languages and functions, Techniques for Turing machine construction, Multi head and Multi tape Turing Machines, The Halting problem, Partial Solvability, Problems about Turing machine- Chomsky hierarchy of languages

Unit V

Decidable and Undecidable Problems in Computability: Unsolvable Problems and Computable Functions, Recursive and recursively enumerable languages, Universal Turing machine, Measuring and classifying complexity - Tractable and Intractable problems, P, NP and NP hard, Decidable and Undecidable problems

SUGGESTED READINGS

1. Hopcroft J.E., Motwani R. and Ullman J.D, "Introduction to Automata Theory, Languages and Computations", Second Edition, Pearson Education.
2. John C Martin, "Introduction to Languages and the Theory of Computation", Third Edition, Tata McGraw Hill Publishing Company, New Delhi
3. Marvin L. Minsky "Computation: Finite and Infinite" – Prentice Hall, 1967
4. Michael Sipser "Introduction to the Theory of Computation", Third Edition, 2012 Cengage Learning
5. Peter Lenz – An Introduction to Formal languages and Automata – 3rd Edition Narosa, 2003
6. Thomas A. Sukamp – An introduction to the theory of computer science languages and machines – 3rd edition, Pearson Education, 2007.
7. G E Reeves - "Introduction to Formal Languages" TMH, 2000.

C. CO-PO & CO-PSO MAPPING TABLE

CO\ PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS O1	PS O2	PS O3
CO1	3	3	2	3	2	0	0	0	2	2	0	3	3	2	0
CO2	2	2	3	2	2	0	0	0	2	2	0	3	2	3	0
CO3	2	2	3	2	2	0	0	0	2	2	0	3	2	3	0
CO4	3	3	3	3	3	1	1	1	3	3	0	3	3	3	1
CO5	3	3	2	3	2	2	2	2	2	2	1	3	3	2	2

D. THEORY LECTURE PLAN

S. No.	CONTENT	NUMBER OF LECTURES	Unit
1	Introduction & Motivation of TOC, Infinite Sets, Closures, Alphabets, Languages & Representation	1	Unit-I (11)
2	Finite Automata: Deterministic FA	1	

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3	Nondeterministic FA	1	
4	Regular expressions	1	
5	Finite Automaton with ϵ - moves	1	
6	Regular Expression, Regular Languages and Kleene's theorem– Conversion of NFA to DFA	1	
7	Equivalence of finite Automaton and regular expressions, Minimization of DFA	2	
8	Arden's Theorem. Myhill Nerode	1	
9	Pumping Lemma for Regular sets, Problems based on Pumping Lemma	2	
CLASS TEST-I			
10	Context Free Grammar: Grammar, Types of Grammar	1	Unit-II (6)
11	Context Free Grammars and Languages, Derivations, Ambiguity	1	
12	Relationship between derivation and derivation trees	1	
13	Elimination of Useless symbols - Unit productions - Null productions	1	
14	Chomsky normal form (CNF),	1	
15	Greibach Normal form (GNF), Problems related to CNF and GNF	1	
16	Pushdown Automata Moves, Instantaneous descriptions, Deterministic/non-deterministic pushdown automata	2	Unit-III (7)
17	Conversion from CFG-to-PDA	1	
18	Conversion from PDA-to-CFG	1	
MID SEMESTER EVALUATION			
19	Equivalence of Pushdown automata and CFL	1	
20	Pumping lemma for CFL, problems based on pumping Lemma	2	

21	Definition and Types of Turing machines, Computable languages and functions	2	Unit-IV (9)
22	Techniques for Turing machine construction	2	
23	Multi head and Multi tape Turing Machines	1	
24	The Halting problem, Partial Solvability	1	
25	Problems about Turing machine	2	
26	Chomsky hierarchy of languages	1	
27	Unsolvable Problems and Computable Functions	1	
28	Recursive and recursively enumerable languages	1	Unit-V (7)
29	Universal Turing machine	1	
30	Measuring and classifying complexity - Tractable and Intractable problems	1	
31	Church-Turing Thesis & Universal Turing machines	1	
32	P, NP and NP-hard problems	1	
33	Decidable and Undecidable problems	1	

E. SELF STUDY

Sr. No.	Topic	Unit
1	(i)How Vedic or philosophical concepts might challenge, inspire, or provide insight into traditional computational theory? (ii)How does the deterministic nature of finite automata (DFA) reflect the Vedic concept of following a clear, preordained path (Svadharma), where every decision is predetermined and fixed?	Unit I
2	(i)What are the foundational principles of "computation" from a Vedic perspective? (ii)In Vedic philosophy, "Brahman" represents an underlying, universal truth. How might context-free grammars (CFG) represent the underlying structure of languages, transcending surface-level variations? (iii)The Vedic concept of "Chitta" (mind) can store infinite possibilities. How does the stack in a pushdown automaton (PDA) symbolize the mind's capacity to process and store infinite possibilities within finite constraints?	Unit II & III

3	(i)Can Vedic mathematical sutras lead to optimal algorithms for complex computations? (ii)How does the concept of undecidability in Turing machines challenge the Vedic understanding of "Maya" (illusion), where some truths may forever remain beyond our grasp?	Unit IV
4	Could Vedic concepts of infinity and cyclic time inform complexity theory?	Unit V

2.2 Computer Architecture and Organization

A. OVERVIEW OF THE COURSE:

Name of Course	Computer Architecture and Organization
Offering Department	COMPUTER SCIENCE & ENGINEERING
B.Tech. Branches to which this course is offered	

OVERVIEW:

The Course on **Computer Architecture and Organization** is offered to the II-year students. This course aims to provide students with a comprehensive understanding of the architecture of processors and the organization of their components, as well as the intricate relationship between hardware and software in digital machines. It covers instruction set architecture (ISA), data path and control unit design, and the functionality of essential CPU components like ALU, register file, and memory. The course delves into memory systems, including cache and virtual memory, and explores input/output organization. Advanced topics such as parallel processing, pipelining, and emerging technologies are also addressed. Additionally, it emphasizes the importance of computation standards and the application of these standards in writing optimized algorithms. By the end of the course, students will appreciate the evolution of computing technology and be equipped to stay abreast of state-of-the-art advancements in the field.

B. SYLLABUS

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COURSE NO	TITLE OF THE COURSE	COURSE STRUCTURE	PRE-REQUISITE
CMCSC402	Computer Architecture and Organization	3L - 1T - 0P	Data Structures
COURSE OUTCOMES (COs) After completion of this course, the students are expected to be able to demonstrate the following knowledge, skills, and attitudes: 1. To understand the architecture of processors and organization of their components, and relationship between hardware and software in digital machines. 2. To design instructions and corresponding logic circuits for a simple CPU with its essential components such as ALU, a register file, memory and input-output. 3. To understand the organization of computer systems 4. To understand the computation standards and using them in writing algorithms 5. To appreciate the evolving technology that governs the evolution of modern computers and continue to keep abreast of state-of-art in computing technology			
COURSE CONTENTS Unit I Basic Structure of Computers: Computer Types, Functional Units, Input Unit, Memory Unit, Arithmetic and Logic Unit, Output Unit, Control Unit, Basic Operational Concepts, Number Representation and Arithmetic Operations, Integers, Floating-Point Numbers, Character Representation, Performance, Technology, Parallelism Historical Perspective. Unit II Instruction Set Architectures: Machine instruction, Machine cycle and Instruction cycles, Instruction Set: memory and non-memory reference instructions, instruction categories: data movement, data manipulation, program control and machine control instructions, CISC types addressing modes and instruction formats, RISC type addressing modes and instruction formats. Unit III Basic Processing Unit Specification of a CPU using RTL, Design of the data path for the simple CPU, Designing the hardwired control path for a CPU, Designing the microprogrammed control path for a CPU. Practical case studies in CISC type and RISC type CPU designs. Unit IV Memory organization: Basic Concepts Semiconductor RAM Memories, Memory Hierarchy, Cache Memories, Direct Memory Access, Performance Considerations. Input output organization: IO interfacing, Asynchronous data transfer, Programmed IO, Interrupt driven IO, Priority schemes, Direct Memory Access, Serial communication techniques Unit V Computer arithmetic: Design of Binary addition and subtraction units, Algorithms for multiplication and division and their implementation, Floating point arithmetic, etc.			

Introduction Parallel Processing and Performance**SUGGESTED READINGS**

1. M. Morris Mano, "Computer System Architecture", PHI
2. J.D. Carpinelli, "Computer Systems Organization and Architecture," Pearson Education
3. William Stallings, "Computer Organization and Architecture, PHI"
4. Heuring and Jordan, Pearson Education, "Computer Systems Design and Architecture"

C. CO-PO & CO-PSO MAPPING TABLE

CO\PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS O1	PS O2	PS O3
CO1	1	1	0	1	1	0	0	0	0	0	0	1	2	3	2
CO2	3	3	3	3	1	0	0	0	0	0	1	1	3	3	1
CO3	1	1	1	1	1	0	0	0	0	0	0	1	2	3	1
CO4	2	1	1	1	1	0	0	0	0	0	0	1	3	2	1
CO5	1	2	1	2	1	1	0	0	0	0	0	3	1	2	3

D. THEORY LECTURE PLAN

S.N o.	CONTENT	NUMBER OF LECTURES	Unit
1	UNIT-I Computer Types, Functional Units	1	Unit-I (7)
2	Input Unit, Output Unit, Memory Unit	1	
3	Arithmetic and Logic Unit, Control Unit	1	
4,5	Basic Operational Concepts, Number Representation and Arithmetic Operations, Integers	2	
6	Floating-Point Numbers, Character Representation	1	
7	Performance, Technology, Parallelism Historical Perspective	1	
8	UNIT-II Machine instruction, Machine cycle and Instruction cycles	1	
CLASS TEST-I			
9,10	Instruction Set: memory and non-memory reference instructions	2	Unit-II

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11, 12	Instruction categories: data movement, data manipulation, program control and machine control instructions	2	(7)
13	CISC types addressing modes and instruction formats	1	
14	RISC type addressing modes and instruction formats	1	
15	UNIT-III Introduction to Basic Processing Unit	1	
16	Specification of a CPU using RTL	1	Unit-III (7)
17	Design of the data path for the simple CPU	1	
18	Designing the hardwired control path for a CPU	1	
MID SEMESTER EVALUATION			
19	Designing the microprogrammed control path for a CPU.	1	
20	Practical case studies in CISC type CPU designs.	1	
21	Practical case studies in RISC type CPU designs.	1	
23	UNIT-IV Introduction to Memory organization	1	Unit-IV (10)
24	Basic Concepts Semiconductor RAM Memories	1	
25	Memory Hierarchy, Cache Memories	1	
26	Direct Memory Access, Performance Considerations	1	
27	Introduction to Input output organization:	1	
28	IO interfacing, Asynchronous data transfer	1	
29	Programmed IO, Interrupt driven IO	1	
30	Priority schemes, Direct Memory Access	1	
31	Serial communication techniques	1	
32	UNIT-V Introduction to Computer arithmetic:	1	
33, 34	Design of Binary addition and subtraction units	2	Unit-V (9)
35, 36	Algorithms for multiplication and division and their implementation	2	
37	Floating point arithmetic	1	
38, 39	Introduction Parallel Processing	2	
40	Performance	1	

E. SELF STUDY

Sr. No.	Topic	
1	Logic Circuits	
2	System-on-a-Chip—A Case Study	
3	The ARM Processor	
4	Examples of Embedded Systems	

2.3 Machine Learning

A. OVERVIEW OF THE COURSE:

Name of Course	Machine Learning
Offering Department	COMPUTER SCIENCE & ENGINEERING

OVERVIEW:

The course on Machine Learning provides students with foundational knowledge and hands-on experience in developing intelligent algorithms that can learn from data. It covers key concepts and techniques, including supervised and unsupervised learning, neural networks, and model evaluation. This course emphasizes practical applications, aiming to prepare students to build and apply machine learning models to solve real-world problems.

B. SYLLABUS

COURSE NO	TITLE OF THE COURSE	COURSE STRUCTURE	PRE-REQUISITE
CMCSC403	Machine Learning	3L - 0T - 2P	Probability and Statistics, Linear Algebra
COURSE OUTCOMES (COs) After completion of this course, students are expected to: - Understand foundational concepts in machine learning, focusing on supervised learning methods. - Implement and evaluate supervised machine learning models using relevant metrics. - Apply neural network models to practical tasks and understand model training techniques. - Project-based learning to Develop machine learning solutions to real-world challenges, interpreting and presenting results effectively.			
COURSE CONTENTS Unit I Introduction to Machine Learning (ML): Introduction to Artificial Intelligence, History and Applications, Definition, and Types of ML: Supervised, Unsupervised, and Reinforcement Learning, ML Workflow and Model Development Cycle, Data Preprocessing: Data Cleaning, Handling Missing Values, Encoding Categorical Data, Data Normalisation, Feature Scaling. Introduction to Python ML Libraries: Scikit-Learn, Pandas, Numpy, Dimensionality Reduction (PCA)			

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Gradient descent, overfitting, underfitting

Unit II

Regression Techniques in Supervised Learning: Linear Regression: Simple and Multiple Linear Regression, Polynomial Regression: Basics and Use Cases, Regularization Techniques: Ridge and Lasso Regression, Feature Engineering, Model Evaluation Metrics for Regression: MSE, MAE, R-squared, RMSE, Bias-Variance Tradeoff.

Unit III

Classification Techniques in Supervised Learning: Classification Basics: Definitions and Applications, Logistic Regression and Decision Boundaries, k-Nearest Neighbors (k-NN), Naïve Bayes, Support Vector Machines (SVM), Decision Trees and ~~Random Forests for Classification~~, ~~Ensemble Learning with Boosting Techniques~~, Model Evaluation Metrics: Accuracy, Precision, Recall, F1 Score, Confusion Matrix, ~~ROC-AUC~~.

Unit IV

Neural Networks and Training Techniques: Introduction to Neural Networks: Perceptron, Activation Functions, Training Neural Networks: Gradient Descent and Ascent, Backpropagation, and Cost Functions, Optimizers, Overfitting and Regularization in Neural Networks, Basics of Deep Learning Architectures: Overview of CNN and RNN

Unit V

Real-world applications, Optimization, and Introduction to Unsupervised Learning: Case Studies in Image Classification, Predictive Analytics, and Recommendation Systems, Model Optimization: Hyperparameter Tuning and Cross-Validation, Introduction to Unsupervised Learning: Clustering, Insights from Industry Applications and Emerging Trends.

SUGGESTED READINGS

1. **Tom Mitchell**, *Machine Learning*, McGraw-Hill, 1997.
2. Aurelien Geron, **Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow**, O'Reilly Media, 2nd Edition, 2019.
3. Ian Goodfellow, Yoshua Bengio, Aaron Courville, **Deep Learning**, MIT Press, 1st Edition, 2016.
4. Peter Flach, **Machine Learning: The Art and Science of Algorithms that Make Sense of Data**, Cambridge University Press, 1st Edition, 2012.
5. Andreas C. Müller, Sarah Guido, **Introduction to Machine Learning with Python: A Guide for Data Scientists**, O'Reilly Media, 1st Edition, 2016.

C. CO-PO & CO-PSO MAPPING TABLE

COPO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS O1	PSO 2	PS O3
CO1	1	3	0	1	3	0	0	0	0	0	0	1	3	2	2
CO2	3	3	2	2	3	0	0	0	0	0	1	1	3	2	2
CO3	3	3	1	2	2	0	0	0	0	0	1	1	2	3	2
CO4	3	3	1	3	2	0	0	0	0	0	2	1	2	2	3

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D.THEORY LECTURE PLAN

S. No.	CONTENT	NUMBER OF LECTURES	Unit
1	Introduction to Artificial Intelligence, History and Applications	1	Unit-I (6)
2	Definition and Types of ML: Supervised, Unsupervised, and Reinforcement Learning	1	
3	ML Workflow and Model Development Cycle	1	
4	Data Preprocessing: Data Cleaning, Handling Missing Values, Encoding Categorical Data, Data Normalisation	2	
5	Introduction to Python ML Libraries: Scikit-Learn, Pandas, Numpy.	1	
6	Linear Regression: Simple and Multiple Linear Regression	1	
7	Polynomial Regression: Basics and Use Cases	1	
8	Regularization Techniques: Ridge and Lasso Regression	1	
9	Feature Engineering, Dimensionality reduction (PCA)	1	
CLASS TEST-I			
10	Model Evaluation Metrics for Regression: MSE, MAE, R-Squared, RMSE, Bias-Variance Tradeoff.	2	Unit-II (6)
11	Classification Basics: Definitions and Applications	1	
12	Logistic Regression and Decision Boundaries	1	
13	k-Nearest Neighbors (k-NN)	1	
14	Naïve Bayes	1	
15	Support Vector Machines (SVM)	1	
16	Decision Trees and Random Forests for Classification	1	
17	Ensemble Learning with Boosting Techniques	2	Unit-III (10)
18	Model Evaluation Metrics: Accuracy, Precision, Recall, F1 Score, Confusion Matrix, ROC-AUC.	2	
MID-SEMESTER EVALUATION			

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24	Introduction to Neural Networks	1	Unit-IV (9)
25	Perceptron, Activation Functions	1	
26	Training Neural Networks: Gradient Descent and Ascent	1	
27	Back-propagation, and Cost Functions	1	
28	Optimizers, Overfitting, and Regularization in Neural Networks	2	
29	Basics of Deep Learning Architectures: Overview of CNN and RNN	1	
30	Expert Topic from Industry	1	
31	Flipped Learning	1	
32	Project Discussion	1	
33	Case Studies in Image Classification	1	Unit-V (9)
34	Predictive Analytics, and Recommendation Systems	1	
35	Model Optimization: Hyperparameter Tuning and Cross-Validation	1	
36	Unsupervised Learning: Clustering	2	
37	Dimensionality Reduction (PCA)	2	
39	Insights from Industry Applications and Emerging Trends.	2	

E. LESSON PLAN for LAB

Lab Class No.	Name of the Experiment	
1	Introduction to Python Lab, Platform, Basic Concepts, List, Tuple, Dictionary, Set.	
2	Python Advance Libraries, Numpy, Pands, Matplotlib, scikit-learn.	
3	Linear Regression	
4	Logistic Regression	
5	Naïve Bayes	
6	SVM	
MID-SEMESTER EVALUATION		
7	Decision Tree & Random Forest	
8	Ensemble learning with boosting technique	
9	Neural Networks	
CLASS TEST-2 (LAB EVALUATION)		
10	Project Based Learning	a. Problem definition for mini project, selecting an ML problem, building an appropriate model
11		b. Evaluate model accuracy, effectiveness, and interpretation of results.
12		c. Mini Project Discussion and Evaluation
END SEMESTER EVALUATION		

F. SELF STUDY

Sr. No.	Topic	
1	Visualizing Gradient Descent: Create a Python script that shows the gradient descent process for a simple linear regression model. Use it to see how the model minimizes errors with each step.	Unit II

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2	Reducing Dimensions with PCA: Use Principal Component Analysis (PCA) on a dataset with many features to reduce it to two dimensions. Visualize the results and interpret how PCA helps simplify data while keeping important information.	Unit II & III & V
3	Vedic Mathematics for Predictive Text Generation: Using the "Ekadhikena Purvena" rule in Vedic mathematics, generate a sequence based on this pattern and develop a simple predictive model to continue the sequence. Analyze how well the model captures the pattern.	Unit V
4	Any topic as suggested by Course instructor	

2.4 Design and Analysis of Algorithms

A. OVERVIEW OF THE COURSE:

Name of Course	Design and Analysis of Algorithms
Offering Department	COMPUTER SCIENCE & ENGINEERING

OVERVIEW:

The course on **Design and Analysis of Algorithms** is offered to second year/third year students and aims to provide an in-depth understanding of the principles and methodologies used to design and analyze algorithms. Through this course, students will learn to evaluate the performance of algorithms using asymptotic notations and understand various computational models and complexity analysis techniques. Students will explore fundamental algorithmic paradigms such as divide and conquer, dynamic programming, greedy methods, and backtracking etc. They will gain practical knowledge in implementing and analyzing sorting and searching algorithms, including advanced data structures like AVL trees, red-black trees, and B-trees. The curriculum also covers essential graph algorithms for tasks such as shortest paths, minimum spanning trees, and topological sorting. Students will delve into the concept of NP-completeness through classic NP-complete problems like satisfiability, Hamiltonian cycle, and the traveling salesman problem. By the end of the course, students will be equipped with the skills to synthesize efficient algorithms for common engineering design situations, analyze their performance, and apply appropriate algorithmic techniques to solve complex problems. This solid foundation in algorithm design and analysis will prepare students for advanced studies and careers in computer science, software development, and related fields.

B. SYLLABUS

COURSE NO	TITLE OF THE COURSE	COURSE STRUCTURE	PRE-REQUISITE
CMCSC404	Design and Analysis of Algorithms	3L - 0T - 2P	Data Structures
COURSE OUTCOMES (COs) After completion of this course, the students are expected to be able to demonstrate the following knowledge, skills, and attitudes: - To be able to analyze the asymptotic performance of algorithms. - To implement and analyze the performance of fundamental data structures. - Students will gain comprehensive knowledge of graph representation and traversal techniques. - To be able to apply important algorithmic design paradigms and methods of analysis. - Students will understand the concepts of P and NP classes, and NP-completeness.			

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COURSE CONTENTS**Unit I**

Analysis of Algorithms: Introduction to RAM model of computation, asymptotic notations and their significance, complexity analysis of algorithms, master's theorem, basic introduction to algorithmic paradigms like divide and conquer, greedy, backtracking, branch and bound.

Unit II

Searching and Sorting: Binary search trees, AVL trees and red-black trees, B-trees, hashing Priority queues, heaps. Sorting: comparison-based sorting - quick sort, heap sort, merge sort: worst and average case analysis. Sorting in linear time - radix sort, bucket sort, counting sort.

String Operations: Boyer-Moore, Knuth-Morris-Pratt (KMP) Algorithm

Unit III

Application of Graph Traversal Techniques: Recall representation of graphs, BFS, DFS, connected components, topological sorting of DAGs, biconnected components, single source shortest path.

Unit IV

Greedy Algorithms: Greedy choice, minimum spanning trees -- Prim's and Kruskal's, Dijkstra's shortest path using arrays and heaps, fractional knapsack, and Huffman coding (use of priority queue).

Dynamic Programming: Binomial Coefficient, matrix chain multiplication, longest common subsequence problem, and 0-1 Knapsack problem,

Unit V

NP-Completeness: P and NP classes, NP-completeness, polynomial time reductions, discussion of different NP-complete problems like satisfiability, Hamiltonian Cycle and travelling salesman problem.

SUGGESTED READINGS

1. T.H. Cormen, C. E. Leiserson, R. L. Rivest "**Introduction to Algorithms**", The MIT Press, 4th Edition, 2022
2. Ellis Horowitz, Sartaj Sahni, Sanguthevar Rajasekaran, "**Computer Algorithms**", Silicon Press, 2nd Edition, 2007.
3. Michael T. Goodrich and Roberto Tamassia, "**Design and Analysis of Algorithms**", Wiley, Edition 2021.
4. Jon Kleinberg And Eva Tardos, "**Algorithm Design**", Pearson, 1st Edition, 2005.

C. CO-PO & CO-PSO MAPPING TABLE

CO\ PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS O1	PS O2	PS O3
CO1	3	3		2	-	2	-	-	-	-	-	1	3	2	-

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CO2	3	2	2	2	-	-	-	-	-	-	-	2	3	2	-
CO3	3	2	2	2	-	-	-	-	-	-	-	2	3	3	1
CO4	3	2	3	2	-	-	-	-	-	-	-	2	3	2	2
CO5	3	2	2	2	-	-	-	-	-	-	-	2	3	2	-

D. THEORY LECTURE PLAN

S. No.	CONTENT	NUMBER OF LECTURES	Unit
1	Introduction to RAM model of computation	1	Unit-I (8)
2	Asymptotic notations and their significance	2	
3	Complexity analysis of algorithms	1	
4	Master’s theorem	2	
5	Basic introduction to algorithmic paradigms like divide and conquer, greedy, backtracking, branch and bound.	2	
CLASS TEST-I			
6	Binary search trees	1	Unit-II (10)
7	AVL trees and red-black trees	1	
8	B-trees	1	
9	Hashing Priority queues, heaps	1	
10	Quick sort, Heap sort	1	
11	Merge sort	1	
12	Sorting in linear time - radix sort, bucket sort	1	
13	Counting sort	1	
14	String operations: Boyer Moore algorithm	1	
15	Knuth-Morris-Pratt Algorithm	1	
Mid-Sem			
14	Recall representation of graphs	1	Unit-III (5)
15	BFS, DFS, connected components, topological sorting of DAGs	2	
16	Biconnected components	1	
17	Single source shortest path	1	
18	Greedy choice	1	Unit-IV (10)
19	Minimum spanning trees -- Prims and Kruskal’s Algorithm	1	
20	Dijkstra’s shortest path using arrays and heaps	1	
21	Fractional knapsack	1	
22	Huffman coding (use of priority queue)	1	
23	Dynamic Programming: Binomial coefficient	1	
24	Matrix chain multiplication	2	
25	Longest common subsequence problem	1	
26	0-1 Knapsack problem	1	

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28	P and NP Classes	1	Unit-V (6)
29	NP-Completeness	1	
30	Polynomial time reductions	1	
31	Discussion of different NP-complete problems like satisfiability, Hamiltonian cycle, travelling salesman problem	3	

E. LESSON PLAN for LAB

Lab Class No.	Name of the Experiment	
1	Sort a given set of elements using the Quicksort method and determine the time required to sort the elements. Repeat the experiment for different values of n, the number of elements in the list to be sorted and plot a graph of the time taken versus n. The elements can be read from a file or can be generated using the random number generator.	
2	Write a program that implements Merge sort sorting methods to sort a given list of integers in ascending order. Compare its running time on input size 10, 100, 1000, 10000 random numbers with quick sort. Show the comparison result graphically.	
3	Find a subset of a given set $S = \{s_1, s_2,...,s_n\}$ of n positive integers whose sum is equal to a given positive integer d. For example, if $S= \{1, 2, 5, 6, 8\}$ and $d = 9$ there are two solutions $\{1,2,6\}$ and $\{1,8\}$. A suitable message is to be displayed if the given problem instance doesn't have a solution.	
4	a. Print all the nodes reachable from a given starting node in a digraph using BFS method. b. Check whether a given graph is connected or not using DFS method.	
5	a. Obtain the Topological ordering of vertices in a given digraph b. Compute the transitive closure of a given directed graph.	
6	Implement 0/1 Knapsack problem using Dynamic Programming	
MID SEMESTER EVALUATION		
7	a. From a given vertex in a weighted connected graph, find shortest paths to other vertices using Dijkstra's algorithm. b. Find Minimum Cost Spanning Tree of a given undirected graph using Prim's algorithm.	
8	Implement N Queen's problem using Back Tracking.	
CLASS TEST-2 (LAB EVALUATION)		
9	Mini Project	a) Study of optimization algorithms and their implementation.
10		b) To apply one nature-inspired optimization algorithm for solving one real world problems
11		c) Modify the working of one optimization algorithm and analyze its computational complexity
12		
END SEMESTER EVALUATION		

F. SELF STUDY

Sr. No.	Topic	
1	Counting of primitive operations	Unit I
2	Hash Tables, Variants of Quick Sort, Parallel Merge Sort	Unit II
3	Bellman-Ford Algorithm, All-pairs shortest paths	Unit IV
4	Vertex-cover, Clique and Set-cover	Unit V
5	Any topic as suggested by Course instructor	

2.5 The course CM ICC 405 **O**ptimization Techniques will be offered by the ICE department. The syllabus for the same will be designed and approved by their respective BOS.